

Every failure mode measured on the VCS has a documented neural-network analogue

The map reads as a necessary-condition screen: on the VCS each failure is proven against the exact §3.2 intervention oracle (full observability, no learning); on neural networks the same failure is a documented analogue, not proven here. A method that fails the screen should not be trusted for stronger claims without further evidence (necessary, not sufficient).

Failure mode measured on the VCS	VCS measured evidence proven in this benchmark	Neural-network analogue documented, not proven here	Status here → there
1 Gradient vanishes on discrete / index outputs	0.000 position-regime F single record Vanilla saliency & IG = 0.000 on the position regime (provably zero at a strobe-timed argmax); whole gradient bucket 0.068.	Shattered / saturated gradients; non-differentiable argmax & index readouts.	proven here analogue there
2 Saliency is model-invariant (sanity-check fails)	0.000 sanity-check margin single record Program-randomization: saliency cosine = 1.000 while 95% of RAM changed -> sanity check FAILS.	Saliency unchanged under model- & label-randomization; behaves like an edge detector.	proven here analogue there
3 Correlation ≠ causation (Granger & tuning trap)	0.136 Granger F 95% CI [0.02, 0.29] Granger-edge faithfulness = 0.136 (false edges vs the true data-flow); game-variable tuning = 0.116.	Spurious Granger edges; 'present ≠ used' tuning & attention read as explanation.	proven here analogue there
4 Single-unit ablation has an interaction blind-spot	0.414 connectome F 95% CI [0.08, 0.78] Single-unit lesion is faithful (0.987) yet misses the wiring: connectome recovered at only 0.414.	Single-unit ablation misleading; computation is distributed across many units.	proven here analogue there
5 SAE / probe: present-not-used + polysemanticity	0.162 selectivity F 95% CI [0.08, 0.24] SAEs match a known variable yet score 0.195 on causal use; probes decode at 0.894 acc but 25 cells are not causal.	Control-task selectivity gap; superposition / polysemantic neurons.	proven here analogue there
6 Baseline-sensitivity of Integrated / Expected Gradients	0.034 EG F (all-regime) 95% CI [0.00, 0.10] IG magnitude is baseline-sensitive (up to 0.999 across games); a content-bytebaseline choice; no canonical baseline kills IG; EG lands at 0.034.	Attribution depends on the (arbitrary) baseline choice; no canonical reference input.	proven here analogue there

■ VCS evidence: proven in this benchmark (vs the exact §3.2 intervention oracle) ■ NN analogue: documented in the literature, not proven by this benchmark

Headline (position regime): causal/intervention 0.412 vs gradient/correlational 0.068 (95% CI 0.390 / 0.070) → a 0.344 faithfulness gap; activation patching = 1.000 (oracle ceiling) where vanilla saliency = 0.000 (naive-gradient floor).